ZAINEB DARCHEM

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LinkedIn | GitHub | Codeforces | Portfolio

OBJECTIVE

Aspiring AI Engineer with strong foundations in applied mathematics, statistics, and data science. Skilled in Python, MLOps, and model development, with hands-on experience in agentic systems, LLM pipelines, NLP, predictive modeling, and recommender systems, with a drive to turn AI into solutions for real-world problems.

EXPERIENCE

Talan Tunisie SummerCamp: AI Intern — July, August 2026

- Helped develop Wink, an AI-powered startup assistance platform built on a multi-agent architecture automating regulatory compliance, taxation, and investment research.
- Built a research-to-startup matching engine using Qdrant, Mistral Embed and LLM-as-a-Judge, connecting researchers with startups to bring scientific research into entrepreneurial projects.
- Developed a copilot for regulatory document analysis and review, with automated regulatory monitoring and generation of reports and pitch decks

Banque Nationale Agricole (BNA) : Data Science Intern — August 2025

- Automated Excel-based accounting workflows using Python, VBA, and Power BI M language.
- Built scripts to combine, clean and transform financial datasets, reducing processing time.
- Integrated automated data preparation directly into Power BI for seamless dashboard refreshes.

EDUCATION

Applied Mathematics and Modeling Engineering — 2024 - 2027
Higher National Engineering School of Tunis (**ENSIT**)

Maths & Physics Preparatory degree : — 2022 - 2024
Preparatory Institute for Engineering Studies of Tunis (**IPEIT**)

SKILLS

- **Programming:** Python, SQL, java, C, C++, MATLAB, HTML/CSS, Javascript
- **AI:** NLP, RAG, Langgraph, Langchain, Transformers, YOLO, SVM, Scikit-Learn, Pandas, TensorFlow, PyTorch
- **Mathematics:** Linear algebra, probability distributions, statistical inference, optimization
- **Tools/Environments:** Docker, Git/GitHub, VSCode, jupyter, google colab

PROJECTS

AutoInsight: LLM-Powered Multi-Agent System for Automated Machine Learning Pipelines (Try it here)

- Built a LangGraph-based multi-agent AutoML system with LLM-powered agents, orchestrated via a central controller to manage end-to-end workflow (EDA, preprocessing, feature engineering, modeling)
- Developed an NLP parser using LLMs to extract user intent (task type) and automatically identify the target column, enabling fully automated pipeline configuration from natural language input.
- Implemented end-to-end automation with validation and reporting, including pipeline consistency checks and generation of structured final reports summarizing data insights, modeling decisions, and performance metrics.

Healthcast : AI-driven Fitness & Nutrition App (Try it here)

- Built an app that converts natural-language health and lifestyle prompts into personalized workout and meal plans.
- Collaborated in a team, sharing roles across NLP application, ML models training and generative AI integration.
- Implemented Random Forest and XGBoost models for workout and nutrition predictions using custom datasets.
- Integrated LLMs to produce motivational podcast-style audio and concise Markdown summaries for users.
- Developed a Streamlit web interface for intuitive user input and output interaction.

COMMUNITY INVOLVEMENTS

- **8th Global Competition on Design for Future Education (GCD4FE) in China**
- **Seneca Hacks :** 48h AI hackathon
- **NerData ENSIT Club :** Project manager organizing hackathons & conducting various AI workshops

ADDITIONAL QUALIFICATIONS

Languages : english (advanced), french (fluent), arabic (native), german (beginner)

Certificates :

- Designing Agentic Systems with LangChain - datacamp
- Mathematics for ML - 365 DataScience
- Hands on TensorFlow - Pistacia